simplest one being randomly choosing some parameters that determines how each edge is drawn - up to, possibly, running some optimization on these parameters to minimize the intersections.

First, however, we need to decide which parameters we are talking about. To keep things simple, we restrict to Bézier curves: quadratic Bézier curves can be described with three parameters (the two endpoints, plus a control point), while the cubic version, which (as shown in figure 16.5) is more flexible, needs two control points, for a total of four 2D points (which make 8 scalar parameters).

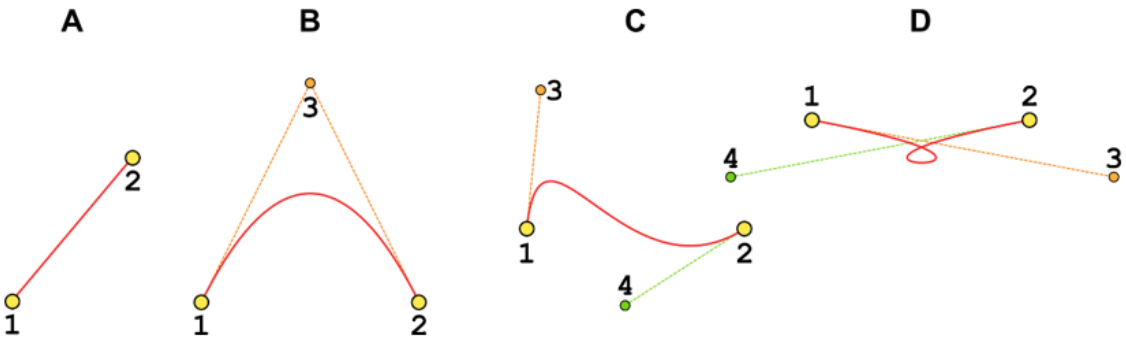


Figure 16.5 Examples of Bézier curves: segments (A) can be considered an edge case with 0 control points, while quadratic curves (B) have 1 control point, and cubic ones (C, D), the most flexible variant, have 2 control points (plus the two endpoints, of course).

We have discussed the details of these representations in the chapter 15, in section 15.4.3: by setting in stone the choice of a subcategory of these curves (the quadratic, symmetric Bézier curves, as shown in figure 16.6), we can already describe how to extend the algorithm in listing 16.1 to deal with the extra parameters.